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Q1. How will a service provider ensure security when they are migrating the services to a cloud?

Ans: Data security should be top priority when working with sensitive information. Cloud data and traffic is often encrypted to ensure its safety. This involves scrambling the data of any given file, meaning it is unreadable to anyone who doesn't have the data encryption key. This ensures that only people with the right data encryption keys can access your data. Traffic between cloud services also passes through an internal network. This means that the traffic should be private to your team and free from outside interference. There are many benefits of migrating to the cloud:-

Scalability: Cloud computing can scale to support larger workloads and more users much more easily than on-premises infra-

Cost: Cloud providers take over maintenance and upgrades, companies migrating to the cloud can spend significantly less on IT operations.

Performance: Migrating to the cloud can improve performance and end-user experience.

Digital Experience: Users can access cloud services and data from anywhere, whether they are employees or customers.

There are some migration strategies who is followed by most of the cloud service provider (CSP):

- i) Rehost: Rehosting or lift and shift involves using infrastructure-as-a-service (IaaS). They simply redeploy their existing data and application on the cloud server. This is easy to do and is thus suited for organizations less familiar with cloud environment.
- ii) Refactor: Refactoring is when they optimize their applications for the cloud. In this case, a platform-as-a-service (PaaS) model is employed. The core architecture of the applications remain unchanged, but adjustments are made to enable the better use of cloud-based tools.

- iii) Revise: Revising builds upon the previous strategies, requiring more significant changes to the architecture and code of the system being moved to cloud.
- iv) Rebuild: Rebuilding takes the revise approach even further by discarding the existing code base and replacing it with a new one.
- v) Replace: Replacing is another solution to the challenges that inform the Rebuild approach. This involves migrating to a third-party, prebuilt application provided by the vendor.

Q2: Explain different service models in cloud computing? Give appropriate example for the same.

Ans: There are three service model in cloud computing :-

i) Software-as-a-service (SaaS)

Software as a service offers applications that are accessed over the web and are not managed by your company, but ^{by} the software provider. This relieves your organisation from the constant pressure of software maintenance,

infrastructure management, network security, data availability and all the other operational issues involved with keeping application up and running. SaaS billing is typically based on factors such as number of users, usage time, amount of data stored and number of transactions processed. This service model has the largest market share in cloud computing. According to Gartner, its sales will reach 117 billion USD by the year 2022. Ex: BigCommerce, Google Apps, Salesforce, Dropbox, Mailchimp, Zendesk etc.

ii) Platform-as-a-service (PaaS)

Platform-as-a-service is halfway between infrastructure-as-a-service (IaaS) and SaaS. It offers access to a cloud-based environment in which users can build and deliver applications without the need of installing and working with IDEs (Integrated Development Environments) which are often very expensive. Additionally, users can often customize the features they want include with subscription. According to Gartner, PaaS has the smallest market share of the

three service models with a project revenue of 27 billion USD by the year 2022. In PaaS users tend to be developers.

EX: AWS Elastic Beanstalk, Heroku, Windows Azure, Openshift, Apache Stratos etc.

iii) Infrastructure-as-a Service (IaaS)

Infrastructure as a service offers a standardized way of acquiring computing capabilities on demand and over the web. Such resources include storage facilities, networks, processing power, and virtual private servers. These are charged under a "pay as you go" model where you are billed by factors such as how much storage you use or the amount of processing power you consume over a certain timespan. In this service model, customers do not need to manage infrastructure, it is up to the provider to guarantee the contracted amount of resources and availability. According to Gartner, this service model is forecasted to grow by 45.9% in 2022.

EX: AWS EC2, Rackspace, Google Compute Engine (GCE), Digital Ocean etc.

Q3. Perform a real-time case study on services provided by any cloud service provider.

Ans: There are many organisations which provide cloud services such as Amazon web services (AWS), Google Cloud Platform (GCP), IBM cloud, Oracle Cloud, and Microsoft Azure.

Case study on Amazon Web Services (AWS)

AWS offers reliable, scalable and inexpensive cloud computing services.

A organisation named, "Siemens" (it is a global powerhouse focusing on the area of electrification, automation and digitalization) which also uses AWS cloud services. Siemens uses an array of AWS services to carry on that tradition of transformation - bringing IIOT to railways and factories, developing intelligent infrastructure for building and distributed energy system, implementing AI into its cybersecurity platform and more. It's never been easier for power plants to monitor their equipment and operations, but the ease of deploying

thousands of sensors results in a challenging volume of sensor data. That's a challenge Siemens facing. The solution Siemens Gas and Power build on AWS has decreased the volume of alerts for its power plant customers by 90 percent. A modern power plant control system receives about 5000 alerts about component failures or out-of-tolerance process conditions a day, but monitoring teams can analyze only 500 a day," says Stefan Lichtenberger, a program manager at Siemens Gas and Power. Because some alerts are less important, while some warn of possible breakdowns or regulatory actions, these teams need help reducing and prioritizing alerts. Project to reduce the volume of alerts can require two full-time employees for six months and because the volume of alerts inevitably rises again - must be repeated every two to three years. From this case study we can understand the importance of cloud computing. Be it any field or any company but the cloud computing is the necessity.